

Quant: Percentage, Ratio and Average

संख्यात्मक अभियोग्यता: प्रतिशत, अनुपात और औसत

What UPSC asks on this - Plain percentage arithmetic from school data: pass percentage, attendance percentage, "x% of N", "what per cent of". - Base traps: "A is 25% more than B, so B is how much less than A?" and a change measured forwards versus backwards. - Successive changes: two increases, two discounts, an increase followed by an equal decrease, and the price-rise versus consumption-cut rule. - Ratio: dividing a grant in a ratio, combining $a : b$ with $b : c$, a ratio that changes when numbers are added, pupil-teacher ratio, partnership shares. - Proportion vocabulary: mean, third and fourth proportional, duplicate and compound ratio. - Averages: a member joins, leaves or is replaced; combined average of two sections; a wrongly copied mark; average speed over equal distances.

How this topic is tested

Topic (iii) questions are one- or two-step arithmetic, and the penalty makes accuracy matter more than speed. The CRT on **01 Nov 2026** is pen-and-paper, **2 hours, 300 marks**, every question carries equal marks, and a wrong answer costs **one-third** of that question's

marks. There is no calculator. The level is that of UPSC's General Ability Test papers, not CSAT.

Three habits save marks here:

- **Assume 100** (or a convenient total such as the LCM) whenever the question gives only percentages or ratios.
- **Use multiplying factors:** +20% is $\times 1.2$ (or $\times 6/5$), -20% is $\times 0.8$ ($\times 4/5$), +12.5% is $\times 9/8$.
- **Check before marking:** a unit digit, a divisibility test or a rough estimate rejects most wrong options in five seconds. A blind guess has zero expected value; once you have eliminated one option, guessing pays.

Percentages: per hundred

A percentage is a fraction with denominator 100, so convert every percent to a fraction first. 'Per cent' literally means 'per hundred'. $x\%$ of $N = N \times x/100$, and 'what per cent is a of b ' = $a/b \times 100$, where b (the word after 'of') is the base.

Fraction-percent equivalents to know cold:

- $1/2 = 50\%$, $1/3 = 33\frac{1}{3}\%$, $1/4 = 25\%$, $1/5 = 20\%$, $1/6 = 16\frac{2}{3}\%$
- $1/7 = 14\frac{2}{7}\%$ (about 14.28), $1/8 = 12.5\%$, $1/9 = 11.11\%$, $1/10 = 10\%$
- $1/11 = 9.09\%$, $1/12 = 8.33\%$, $1/15 = 6.67\%$, $1/16 = 6.25\%$, $1/20 = 5\%$, $1/25 = 4\%$
- Multiples: $3/8 = 37.5\%$, $5/8 = 62.5\%$, $7/8 = 87.5\%$, $2/3 = 66\frac{2}{3}\%$, $3/4 = 75\%$, $4/5 = 80\%$, $5/6 = 83\frac{1}{3}\%$

A handy symmetry: **$x\%$ of $y = y\%$ of x** . So 8% of 25 = 25% of 8 = 2, and 16% of 75 = 75% of 16 = 12.

Example 1 (pass percentage). In a Delhi government school **1,250** students appeared in Class X and **1,075** passed. Pass % = $1,075/1,250 \times 100$. Long method: divide numerator and denominator by 25 to get $43/50 = 86/100 = \mathbf{86\%}$. Shortcut: 1% of 1,250 is 12.5, and $1,075 \div 12.5 = 86$.

Example 2 (working backwards). After a 30% rise, enrolment became 1,040. Original enrolment $\times 1.3 = 1,040$, so original = $1,040/1.3 = \mathbf{800}$. The trap option is $1,040 \times 0.7 = 728$, which wrongly takes 30% of the new value.

Percentage change and the base rule

Percentage change is always measured on the original value, and 'more than B' makes B the base.

% change = $(\text{new} - \text{old})/\text{old} \times 100$. The same absolute change gives a different percentage in each direction because the base changes.

Example 3 (forwards and backwards). Enrolment rose from 840 to 966. Rise = 126; $126/840 = 0.15$, so **+15%**. If it later falls from 966 back to 840, the fall is $126/966 = 3/23 = \text{about } \mathbf{13\%}$, not 15%. Same 126 students, different base.

Example 4 (more versus less). A's salary is 25% more than B's. Let $B = 100$, so $A = 125$. B is 25 less than A on a base of 125: $25/125 = \mathbf{20\% \text{ less}}$. Formula: if A is $x\%$ more than B, B is $\mathbf{100x/(100 + x)\%}$ less; if A is $x\%$ less, B is $\mathbf{100x/(100 - x)\%}$ more.

Fixed pairs worth memorising:

A is more by	B is less by
20% (1/5)	16 2/3% (1/6)
25% (1/4)	20% (1/5)
33 1/3% (1/3)	25% (1/4)
50% (1/2)	33 1/3% (1/3)
100% (1/1)	50% (1/2)

Memory aid: **$1/n$ more is $1/(n + 1)$ less.** The denominator climbs by one when you switch direction.

Percentage points are not per cent. If a school's pass rate goes from 80% to 90%, it has risen by **10 percentage points** but by $10/80 = 12.5$ per cent. UPSC options often offer both numbers.

Successive percentage changes

Apply the multiplying factors one after another; never add the percentages. For changes of $x\%$ then $y\%$, the net change is $x + y + xy/100$ (keep the signs). Two equal and opposite changes always leave a loss of $x^2/100 \%$.

Example 5 (two increases). A fee of Rs 2,000 is raised by 10% and then by 20%. Long method: 2,000 becomes 2,200, then 2,640, a rise of 640 on 2,000 = **32%**.

Shortcut: $10 + 20 + (10 \times 20)/100 = 32$. Adding to get 30% is the trap.

More patterns from the same formula:

- Discounts of 20% and 10%: $-20 - 10 + 2 = -28\%$, not 30%.

- +10% then -10%: $10 - 10 - 1 = -1\%$. +20% then -20%: -4% .
- +20% then +20%: $20 + 20 + 4 = +44\%$ (this is why 2 years of compound growth at 20% is 44%).

Example 6 (price rise versus consumption). Stationery prices rise by 20%. By what per cent must a school cut consumption to keep the same budget? Expenditure = price $\times$ quantity. The new price factor is $6/5$, so the quantity factor must be $5/6$: a cut of $1/6 = 16\frac{2}{3}\%$.

Formula: cut = $100x/(100 + x)\%$. The matching rule for a price fall of $x\%$: consumption can rise by $100x/(100 - x)\%$ (a 20% fall allows a 25% rise). Answering 20% is the trap.

Growth over several years

Repeated percentage growth compounds: final = $P(1 + r/100)^n$, and a repeated fall is $P(1 - r/100)^n$. This is the same arithmetic as compound interest and population growth.

Example 7 (enrolment projection). Enrolment of 2,000 grows 10% a year for 2 years: $2,000 \times 1.1 \times 1.1 = 2,000 \times 1.21 = 2,420$. If it instead falls 10% a year for 2 years: $2,000 \times 0.81 = 1,620$. The second answer is not 2,000 - 400, because the second 10% is taken on 1,800.

School example. Result analysis in a Delhi government school reports pass %, the share of students scoring 60% and above, and section-wise comparisons. A Vice Principal who knows that $1/8$ is 12.5% reads "45 of 360 students" as 12.5% without a calculator, and sees that

a jump from 10 to 15 distinctions is a 50% rise even though it is only 5 students.

Ratio: comparing two quantities

A ratio compares two quantities in the same unit, has no unit itself, and does not change when both terms are multiplied or divided by the same number.

Write $a : b$ for a/b , always simplify, and keep the order the question uses. A ratio of 3 : 2 for boys to girls says nothing about actual numbers until a total or one quantity is given.

Two facts that generate questions:

- **Dividing a sum in a ratio:** share = (own part/total parts) $\times$ sum.
- **Adding the same number to both terms** pushes the ratio towards 1 : 1; multiplying both terms leaves it unchanged. (3 : 2 becomes 13 : 12 when 10 is added to each term; it stays 3 : 2 when each term is doubled.)

Example 8 (splitting a grant). A composite grant of **Rs 75,000** is to be divided among library, sports and laboratory in the ratio 2 : 3 : 5. Total parts = 10, so one part = Rs 7,500. Shares: **15,000, 22,500, 37,500**. Check: they add back to 75,000.

Combining two ratios

When two ratios share a term, make that term equal in both and read off the three-term ratio.

Example 9. $a : b = 3 : 4$ and $b : c = 5 : 6$. The common term b is 4 in one ratio and 5 in the other; LCM = 20.

Multiply the first ratio by 5 ($15 : 20$) and the second by 4 ($20 : 24$). So $a : b : c = 15 : 20 : 24$ and $a : c = 15 : 24 = 5 : 8$. Shortcut for $a : c$ alone: multiply straight across, $(3 \times 5) : (4 \times 6) = 15 : 24$.

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`mermaid flowchart TD
  A[a:b = 3:4] --> C[Make b equal, LCM 20]
  B[b:c = 5:6] --> C
  C --> D[a:b = 15:20]
  C --> E[b:c = 20:24]
  D --> F[a:b:c = 15:20:24]
  E --> F
```

A ratio that changes

Set the two quantities as $3x$ and $2x$, write the new ratio as an equation, and solve for x . This handles ages, boys-and-girls, and income-expenditure questions.

Example 10 (students join). Boys and girls in a school are in the ratio $3 : 2$. If 10 more girls are admitted the ratio becomes $6 : 5$. Let boys = $3x$ and girls = $2x$. Then $3x/(2x + 10) = 6/5$, so $15x = 12x + 60$ and $x = 20$. Boys = 60, girls = 40, so the school had **100** students before the admissions. Check: $60 : 40 = 3 : 2$.

Proportion vocabulary

Four numbers are in proportion when $a : b = c : d$, which means the product of the extremes equals the product of the means ($ad = bc$). UPSC asks the named special cases, so learn the formulas as a set.

Term	Formula	Example
Fourth proportional to a, b, c	bc/a	4, 6, 10 gives 15
Third proportional to a, b	b^2/a	8, 12 gives 18
Mean proportional between a and b	square root of ab	4 and 25 gives 10
Duplicate ratio of a : b	$a^2 : b^2$	2 : 3 gives 4 : 9
Sub-duplicate ratio	root a : root b	16 : 25 gives 4 : 5
Triplicate ratio	$a^3 : b^3$	2 : 3 gives 8 : 27
Compound ratio	multiply term by term	2 : 3 and 4 : 5 give 8 : 15

Example 11 (all three at once). Fourth proportional to 4, 6, 10: $4/6 = 10/x$, so $4x = 60$ and $x = 15$. Third proportional to 8 and 12: $8/12 = 12/x$, so $x = 144/8 = 18$. Mean proportional between 4 and 25: square root of $(4 \times 25) = \text{square root of } 100 = 10$. Memory aid: **'third' uses a square, 'mean' uses a square root.**

Continued proportion $a : b = b : c$ is the same statement as 'b is the mean proportional between a and c'.

Direct and inverse proportion, pupil-teacher ratio, partnership

Direct proportion keeps the ratio y/x constant; inverse proportion keeps the product xy constant.

More students need proportionately more textbooks (direct). More teachers mean fewer students per teacher (inverse).

Example 12 (pupil-teacher ratio). A school has 1,800 students and 50 teachers. $PTR = 1,800/50 = 36 : 1$. To bring it down to $30 : 1$ the school needs $1,800/30 = 60$ teachers, so **10 more**. The RTE norms are usually quoted as **30 : 1** for Classes I to V and **35 : 1** for Classes VI to VIII. Strictly, the RTE Schedule fixes primary teachers by enrolment slabs (2 teachers for up to 60 children, i.e. $30 : 1$) and allows up to $40 : 1$, excluding the head teacher, where enrolment is above 200; for upper primary it requires at least one teacher for every 35 children. The Act is covered in the law pack, but UPSC can combine the norm with this arithmetic.

Example 13 (partnership). Profit shares are in the ratio of **capital $\times$ time**. A invests Rs 50,000 for 12 months and B invests Rs 30,000 for 8 months. Ratio = $(50,000 \times 12) : (30,000 \times 8) = 600 : 240 = 5 : 2$. A profit of Rs 35,000 is split into 7 parts of 5,000: A gets **25,000**, B gets **10,000**. Ignoring time and splitting $5 : 3$ is the trap.

Averages: total divided by count

Every average question is a total question in disguise, so write total = average $\times$ count before anything else. Average (arithmetic mean) = sum of observations / number of observations.

Properties that answer questions without calculation:

- Add k to every value: the average rises by exactly k .
Multiply every value by k : the average is multiplied by k .

- Average of consecutive numbers, or of any arithmetic progression, = **(first + last)/2** = the middle term.
- Average of the first n natural numbers = **$(n + 1)/2$** ; of the first n even numbers = **$n + 1$** ; of the first n odd numbers = **n** .
- The average always lies between the smallest and the largest value.

Example 14 (sequences). Average of 1 to 50 = $51/2 = 25.5$. Average of all odd numbers from 1 to 99 = $(1 + 99)/2 = 50$. Average of the multiples of 5 from 5 to 100 = $(5 + 100)/2 = 52.5$.

Deviation shortcut for raw marks. Average of 62, 65, 58, 71, 69: assume 65; the deviations are -3, 0, -7, +6, +4, which sum to 0, so the average is exactly **65**. Adding five numbers and dividing takes longer and invites slips.

```
`mermaid flowchart TD
  A[Old average x count] --> B[Old total]
  B --> C{What changed?}
  C --> D[A value joins]
  C --> E[A value leaves]
  C --> F[A value is replaced]
  D --> G[New total / new count]
  E --> G
  F --> G
  style A fill:none,stroke:none
  style B fill:none,stroke:none
  style C fill:none,stroke:none
  style D fill:none,stroke:none
  style E fill:none,stroke:none
  style F fill:none,stroke:none
  style G fill:none,stroke:none`
```

Someone joins, leaves, or is replaced

Only the total and the count change; track those two numbers and the new average follows.

- Replacement (count unchanged): new average = old average + **$(\text{new value} - \text{old value})/n$** .
- A newcomer: newcomer's value = new average + **old count $\times$ rise in average**.

- A wrongly copied value: correct average = wrong average + $(\text{correct} - \text{wrong})/n$.

Example 15 (replacement). The average age of 5 teachers is 40. A teacher aged 50 retires and one aged 30 joins. Long method: total $200 - 50 + 30 = 180$, and $180/5 = 36$. Shortcut: $40 + (30 - 50)/5 = 40 - 4 = 36$.

Example 16 (newcomer). The average mark of 9 students is 72. A tenth student joins and the average becomes 73. Long method: $10 \times 73 - 9 \times 72 = 730 - 648 = 82$. Shortcut: the newcomer brings 73 for themselves plus 1 extra mark for each of the 9 others: $73 + 9 = 82$.

Example 17 (wrong entry). The average of 20 marks was calculated as 35, but a mark of 82 had been entered as 28. Correct average = $35 + (82 - 28)/20 = 35 + 2.7 = 37.7$. Dividing the 54 by 10, or forgetting to divide at all, are the usual traps.

Weighted average of two groups

When two groups of different sizes are combined, weight each average by its group size; the combined average sits nearer the larger group. Combined average = $(n_1 \times x_1 + n_2 \times x_2)/(n_1 + n_2)$. Only when the groups are equal in size is it the simple mean of the two averages.

Example 18 (two sections). Section A (40 students) averages 65 and Section B (35 students) averages 50. Combined = $(40 \times 65 + 35 \times 50)/75 = (2,600 + 1,750)/75 = 4,350/75 = 58$. The simple mean 57.5 is the trap. Check by alligation: the group sizes are in the ratio of

the distances from the combined mean, $(58 - 50) : (65 - 58) = 8 : 7 = 40 : 35$.

Average speed

For equal distances at speeds x and y , the average speed is $2xy/(x + y)$, never $(x + y)/2$. Time, not distance, is what gets weighted: the slower leg takes longer, so it pulls the average down. For equal times, the average speed is the simple mean.

Example 19. A teacher drives to school at 40 km/h and returns the same way at 60 km/h. Average speed = $2 \times 40 \times 60 / (40 + 60) = 4,800 / 100 = \mathbf{48 \text{ km/h}}$. Long method with a 120 km leg: 3 hours out, 2 hours back, 240 km in 5 hours = 48 km/h.

Common traps

- **Wrong base.** '25% more' flips to '20% less'; the fall from 966 to 840 is 13%, not 15%. Always ask: per cent of what?
- **Adding successive percentages.** 10% then 20% is 32%; discounts of 20% and 10% are 28%; +10% then -10% is -1%.
- **Price-rise equals consumption-cut.** A 25% rise needs a 20% cut, a 20% rise needs a 16 2/3% cut.
- **Percentage versus percentage points.** 80% to 90% is 10 points and 12.5 per cent.
- **Reverse percentage.** 'After a 30% rise it is 1,040' means divide by 1.3, not multiply by 0.7.
- **Compound fall.** Two falls of 10% leave 81%, not 80%.

- **Ratio order.** 3 : 2 and 2 : 3 are different answers; the combined ratio is 15 : 20 : 24, not 3 : 4 : 6.
- **Partnership without time.** Shares follow capital $\times$ time, not capital alone.
- **Third versus mean proportional.** Third proportional to 8, 12 is 18 (square); mean proportional between 4 and 25 is 10 (square root).
- **Simple mean of two sections.** Weight by size: 58, not 57.5.
- **Average speed.** 48 km/h, not 50 km/h, for 40 and 60 over equal distances.
- **Wrong-entry average.** Divide the correction by the full count: $54/20 = 2.7$, not $54/10$.

Remember for the exam

1. $x\%$ of $N = N \times x/100$; $x\%$ of $y = y\%$ of x .
2. $1/6 = 16\frac{2}{3}\%$, $1/7$ is about 14.28%, $1/8 = 12.5\%$, $1/9 = 11.11\%$, $1/11 = 9.09\%$, $1/12 = 8.33\%$, $1/16 = 6.25\%$.
3. $\%$ change = $\text{change}/\text{original} \times 100$; the base is the earlier value or the value after 'than'.
4. $x\%$ more means $100x/(100 + x)\%$ less; 25% more = 20% less; $1/n$ more is $1/(n + 1)$ less.
5. Successive $x\%$ and $y\%$: net = $x + y + xy/100$; equal up and down = loss of $x^2/100\%$.
6. Price up $x\%$: cut consumption by $100x/(100 + x)\%$; price down $x\%$: raise it by $100x/(100 - x)\%$.
7. n years of $r\%$ growth: $P(1 + r/100)^n$; 10% for 2 years = 21%, 20% for 2 years = 44%.
8. Share in a ratio = $\text{own part}/\text{total parts} \times \text{sum}$; combine $a : b$ and $b : c$ by equalising b (LCM).

9. $a : b = c : d$ means $ad = bc$; fourth proportional bc/a ; third proportional b^2/a ; mean proportional = square root of ab .
10. Duplicate ratio $a^2 : b^2$; compound ratio multiplies term by term.
11. Partnership profit ratio = capital $\times$ time.
12. Total = average $\times$ count; add k to all and the average rises by k .
13. Average of an AP = $(\text{first} + \text{last})/2$; first n natural numbers $(n + 1)/2$; first n odd numbers n ; first n even numbers $n + 1$.
14. Replacement: new average = old + $(\text{new} - \text{old})/n$; newcomer = new average + old count $\times$ rise.
15. Weighted average = $(n_1 \times x_1 + n_2 \times x_2)/(n_1 + n_2)$; average speed over equal distances = $2xy/(x + y)$.

Sources

- UPSC Special Advertisement No. 51/2026, Appendix-I: scheme and syllabus of the Combined Recruitment Test (2026).
- UPSC Recruitment Test question paper and answer key, Vice Principal, DoE GNCTD, Advt No. 10/2022 (pattern reference).
- NCERT Mathematics (editions before the new Ganita Prakash series): Class 6, 'Ratio and Proportion'; Classes 7 and 8, 'Comparing Quantities'; Class 7, 'Data Handling' (arithmetic mean).
- RTE Act 2009, Section 25 and the Schedule (pupil-teacher norms).

- R.S. Aggarwal, Quantitative Aptitude for Competitive Examinations, S. Chand (percentage, ratio and proportion, average, partnership).
- Unit notes: units/reason.md, sections 13 to 15 (2026).

Quick facts

- $x\%$ of $N = N \times x/100$, and $x\%$ of $y = y\%$ of x (8% of $25 = 25\%$ of $8 = 2$).
- $1/6 = 16\frac{2}{3}\%$, $1/7 \approx 14.28\%$, $1/8 = 12.5\%$, $1/9 = 11.11\%$, $1/11 = 9.09\%$, $1/12 = 8.33\%$, $1/16 = 6.25\%$.
- Percentage change is always on the original value; 'A is 25% more than B' means B is 20% less than A ($100x/(100 + x)$).
- Successive changes of $x\%$ and $y\%$: net = $x + y + xy/100$; $+10\%$ then $+20\% = 32\%$, discounts 20% and $10\% = 28\%$, $+10\%$ then $-10\% = -1\%$.
- Price up $x\%$: cut consumption by $100x/(100 + x)\%$ (25% rise needs a 20% cut; 20% rise needs $16\frac{2}{3}\%$).
- Growth of $r\%$ for n years compounds: $P(1 + r/100)^n$; two falls of 10% leave 81% , not 80% .
- 80% to 90% is a rise of 10 percentage points but 12.5 per cent.
- Share in a ratio = (own part/total parts) $\times$ sum; Rs 75,000 in $2 : 3 : 5$ gives 15,000, 22,500, 37,500.

- Combine $a : b = 3 : 4$ and $b : c = 5 : 6$ by equalising b (LCM 20): $a : b : c = 15 : 20 : 24$.
- $a : b = c : d$ means $ad = bc$; fourth proportional = bc/a , third proportional = b^2/a , mean proportional = square root of ab .
- Partnership profit is shared in the ratio capital $\times$ time; Rs 50,000 for 12 months and Rs 30,000 for 8 months give 5 : 2.
- Total = average $\times$ count; adding k to every value raises the average by exactly k .
- Average of an arithmetic progression = (first + last)/2; average of 1 to $n = (n + 1)/2$.
- Replacement: new average = old + (new - old)/ n ; a wrongly copied value: correct average = wrong average + (correct - wrong)/ n .
- Weighted average = $(n_1 \times x_1 + n_2 \times x_2)/(n_1 + n_2)$; average speed over equal distances = $2xy/(x + y)$, so 40 and 60 km/h give 48 km/h.